REVIEW

This review only gives you an idea of what the test is like and helps you identify areas you need to review before test day.

Find a proposition that is logically equivalent to

$$(p \rightarrow q) \wedge (p \rightarrow \neg q)$$

(i) $\neg$q (ii) F (iii) $\neg$p (iv) T

Let $F(x)$ = "x is a student of the Bussiness department",

$J(x)$ = "x knows the computer language *Java*",

$R(x)$ = "x is can speak *Russian*".

The domain consists of all students of the university.

Express the sentence into a logical expression

"Some student of the Bussiness department either can speak *Russian*, or knows *Java*"

(i) $\forall x(F(x) \rightarrow (J(x) \vee R(x)))$

(ii) $\forall x(F(x) \rightarrow (J(x) \wedge R(x)))$

(iii) $\exists x(F(x) \wedge (J(x) \vee R(x)))$

(iv) $\exists x(F(x) \vee (J(x) \vee R(x)))$

Let F(x) = "x is a student of the Business department"
J(x) = "x knows the computer language Java"
R(x) = " x is can speak Russian".
Translate the sentence into logical expression
"There is a student of the Business department who can speak Russian, but does not know Java"

Let p, q be the propositions

p = "She is out of work"
q = "She spends less on clothes"

Translate the sentence into logical expression

"*Although she is out of work, she doesn't spend less on clothes.*"

(i) $p \rightarrow \neg q$

(ii) $p \wedge \neg q$

(iii) $p \vee \neg q$

(iv) $p \leftrightarrow \neg q$

(v) $\neg q \rightarrow p$

Let p and q be the propositions
p :You drive over 65 miles per hour.
q :You get a speeding ticket.
Write the proposition using p and q and logical connectives (including negations).
"Whenever you get a speeding ticket, you are driving over 65 miles per hour".

A) $q \leftrightarrow p$

B) $\neg q \leftrightarrow p$

C) $q \rightarrow p$

D) $q \vee \neg p$

Choose a tuple of values of p, q, r, s such that the proposition

$$(p \rightarrow q) \wedge (r \rightarrow s)$$

is False.

a) p=q=r=F, s=T
b) P=q=T, r=s=F
c) P=r=s=F, q=T
d) P=s=T, q=r=F

Which argument is valid?

A. Any freshman must take Discrete Mathematics. An is a freshman. Therefore, An must take Discrete Mathematics.

B. Any student has a laptop. An has a laptop. Therefore, An is a student.

C. Any student can access this website. An is not a student. Therefore, An can not access this website.

D. Any one who score below 4 in the exam will have to retake the class. An retook this class. Therefore An scored below 4 in the exam.

Let Q(x, y) = "x+y=0" on the set of integers. Which proposition is TRUE?

(i) $\exists x \forall y Q(x, y)$

(ii) $\forall x \exists y Q(x, y)$

(iii) $\forall x \forall y Q(x, y)$

(iv) $\exists y \forall x Q(x, y)$

Given the universal set U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}.

Represent the subset {1, 2, 5, 7, 8, 10} by a bit string whose i-th bit is 1 if i belongs to that subset, and is 0 otherwise. What is that

A. 0011010010

B. 1001011001

C. 0110100110

D. None of the other choices is correct

E. 1100101101

Which of the following is a function?

(i) $f: \mathbf{N} \to \mathbf{N}, \quad f(n) = \sqrt{n}$

(ii) $f: \mathbf{N} \to \mathbf{R}, \quad f(n) = \sqrt{n}$

(iii) $f: \mathbf{R} \to \mathbf{N}, \quad f(n) = \sqrt{n}$

(iv) $f: \mathbf{R} \to \mathbf{R}, \quad f(n) = \sqrt{n}$

Which of the following functions $f: R \rightarrow R$ is NOT one-to-one?

(i) $f(x) = 3x + 21$

(ii) $f(x) = 2x^3 + 5$

(iii) $f(x) = \lfloor 2x \rfloor$

Let N = {0, 1, 2, 3, ….}.

Which of the following functions are onto from N to N?

$f(n) = \left\lceil n - \frac{1}{2} \right\rceil$

$g(n) = \lfloor n \rfloor$

$h(n) = n+1$

Let $f: \mathbf{Z} \to \mathbf{N}$ be the function defined as follows

$$f(n) = \begin{cases} -2n, & n \leq 0 \\ 2n-1, & n > 0 \end{cases}$$

Choose the correct answer.

A. f is both one-to-one and onto.

B. f is onto but not one-to-one.

C. f is neither one-to-one nor onto.

D. f is one-to-one but not onto.

Find the number of positive integers that do not exceed 2021 and are divisible by 7 or 9.

Find the number of positive integers that do not exceed 100 and are divisible by 3 or 4.

How many primes are in {89. 111. 103. 205}?

A. None of the other choices is correct

B. 3

C. 4

D. 1

E. 2

Determine whether the integers in each of these sets are pairwise relatively prime.

a) 21, 34, 55

b) 14, 17, 85

c) 25, 41, 49, 64

d) 17, 18, 19, 23

Find gcd(21!,12!)

A. 12

B. 120

C. 21!

D. 12!

D.

If the product of two integers is $2^7 3^8 5^2 7^4$ and their least common multiple is $2^4 3^4 5^1 7^2$, then their greatest common divisor is _____

A person deposited 10 millions VND in a saving account at the rate of 12% a year. After 10 years, how much money will be in the account?

Encrypt the message PASS by translating the letters into numbers, applying the given encryption function, and then translating the numbers back into letters. a) f(p) = (p + 3) mod 26 (the Caesar cipher) b) f(p) = (3p + 7) mod 26.

A pseudorandom number sequence is generated as follows:

$$x_0 = 2, \quad x_n = (3x_{n-1} + 2) \bmod 11.$$

Find x_3.

Suppose that $f(x) = x^2 \log x$ and $g(x) = x \log x^4$. Which statement is True?

(i) $f(x) = O(g(x))$ (ii) $g(x) = O(f(x))$

ii

Suppose that $f(x) = x^2 \log x$ and $g(x) = (x \log x^4)^2$. Which statement is True?

(i) $f(x) = O(g(x))$ (ii) $g(x) = O(f(x))$

Let $f(x) = x^2\log(x)$ and $g(x) = x\log(x^2)$.

Choose the correct statements:

(i) $f(x) = O(g(x))$

(ii) $g(x) = O(f(x))$

Convert $(1110011100)_2$ to octal base.

Let $X = \{\emptyset, \{\emptyset\}, \{a\}, \{b, c\}\}$ and $Y = \{x, \{y, y\}, x\}$. How many one-to-one functions are there from Y to X?

A. 4 B. 5 C. 6 D. 10 E. 12

Number of one-to-one functions from a set with m elements to one with n elements ($n \geq m$):

$$P(n, m) = \frac{n!}{(n-m)!}$$

C

Use the greedy algorithm to make change using quarters, dimes, nickels, and pennies for a) 63 cents b) 89 cents.

What is total number of coins used?

Which integer is congruent to 5 mod 17

A. 103 B. 80 C. -122 D. -29

Find the smallest integer a such that for all integer $n \geq a$ we can write $n = 5x + 7y$ where x and y are non-negative integers

A. 19 B. 31 C. 12 D. 24

Find the output of the recursive algorithm if the input n = 5

Procedure ORD(n:int)
If n = 1 then f(1) = 3
Else f(n):=f(n-1)*n

Given a sequence $\{a_n\}$ satisfying the recurrence relation

$$a_0 = -1,\ a_n = a_{n-1} + 2^n \text{ for } n = 1, 2, \ldots$$

Find a_6.

Give a recursive definition of the sequence

$\{a_n=5n\}, n = 1, 2, 3, \ldots$

(i) $a_0=0, a_n= a_{n-1}+5$ for $n = 1, 2, 3, \ldots$

(ii) $a_1=1, a_n= a_{n-1}+5$ for $n = 2, 3, \ldots$

(iii) $a_1=5, a_{n-1}= a_n+5$ for $n = 2, 3, \ldots$

(iv) $a_1=5, a_n= a_{n-1}+5$ for $n = 2, 3, \ldots$

Give a recursive definition of the sequence $\{an\}$, n=1, 2, 3, . . . if $a_n = 6n$.

(A) $a_{n+1} = a_n + 6 \ for \ n \geq 1 \ and \ a_1 = 6$

(B) $a_{n+1} = a_n - 6 \ for \ n \geq 1 \ and \ a_1 = 6$

(C) $a_{n+1} = 6a_n \ for \ n \geq 1 \ and \ a_1 = 6$

(D) $a_{n+1} = a_n + 6 \ for \ n \geq 1 \ and \ a_1 = 1$

Find a recursive definition for A={(x,y): x, y are positive integers and x+y is even}

(a) $(1,1) \in A$. If $(x, y) \in A$ then $(x, y + 2) \in A$ and $(x + 2, y) \in A$
(b) $(1,1) \in A$. If $(x, y) \in A$ then $(\mathrm{x} + 1, \mathrm{y} + 1) \in A, (x, y + 2) \in A$ and $(x + 2, y) \in A$
(c) $(1,1) \in A$. If $(x, y) \in A$ then $(\mathrm{x} + 1, \mathrm{y} + 1) \in A$
(d) $(1,1) \in A$. If $(x, y) \in A$ then $(\mathrm{x} + 2, \mathrm{y} + 2) \in A$

A 12 vertices graph G has 2 vertices of degree 3, remaining vertices of degree 4. How many edges are there in this graph?

A. 19 B. 20 C. 21 D.22 E.23

THE HANDSHAKING THEOREM Let $G = (V, E)$ be an undirected graph with m edges. Then

$$2m = \sum_{v \in V} \deg(v).$$

E

The **complementary graph** $\overline{G}$ of a simple graph G has the same vertices as G. Two vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G.

62. If G is a simple graph with 15 edges and $\overline{G}$ has 13 edges, how many vertices does G have?

63. If the simple graph G has v vertices and e edges, how many edges does $\overline{G}$ have?

Which of the following graphs has no Euler circuit?

C_4, Q_4, W_4, K_5

A. C_4 B. K_5 C. Q_4 and K_5 D. W_4

D

List all positive integers *n* such that W_n has a Hamilton circuit?

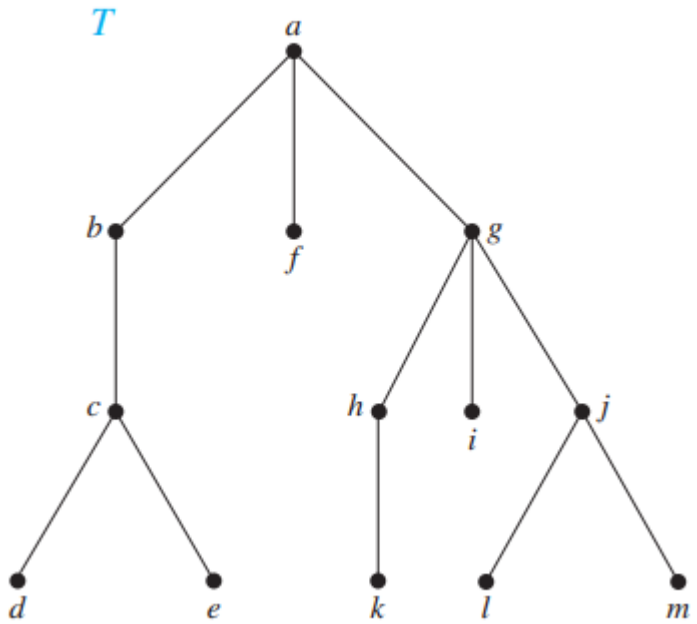
A. 2 B. 4 C. 5 D. all positive integers n>2

D

Which vertices are ancestors of

a) i ?

b) f ?



How many leaves does a full 3-ary tree with 40 vertices have

A. 13 B. 81 C. 27 D. 121

A full m-ary tree with n vertices has $i = (n - 1)/m$ internal vertices and $l = [(m - 1)n + 1]/$ m leaves

Which integer is congruent to 5 mod 17?

A)103

B)80

C)29

D) -122

Let 131 div 17 = x, and -131 div 17 =y. Find x + y?

A)2
B)1
C)0
D)-1

Let T be a full 3-ary tree such that every leaf is at level 4. How many vertices does T have?

ALGORITHM 5 The Insertion Sort.

procedure *insertion sort*($a_1, a_2, \ldots, a_n$: real numbers with $n \geq 2$)
for $j := 2$ **to** n
    $i := 1$
    while $a_j > a_i$
        $i := i + 1$
    $m := a_j$
    for $k := 0$ **to** $j - i - 1$
        $a_{j-k} := a_{j-k-1}$
    $a_i := m$
{$a_1, \ldots, a_n$ is in increasing order}

Input data: 3, 2, 4, 7, 1, 6, 5
What is the order of the elements in the list after running the outer loop with
a) i=4
b) i=5

ALGORITHM 4 The Bubble Sort.

procedure *bubblesort*($a_1, \ldots, a_n$: real numbers with $n \geq 2$)
for $i := 1$ **to** $n - 1$
    for $j := 1$ **to** $n - i$
        if $a_j > a_{j+1}$ **then** interchange a_j and a_{j+1}
{$a_1, \ldots, a_n$ is in increasing order}

Input data: 3, 2, 4, 7, 1, 6, 5
What is the order of the elements in the list after running the outer loop with

a) i=4
b) i=5

Which of the following procedures are recursive algorithms?

(A)Procedure ORD(x: int, n: positive int)

```
d:=0
for i = 1 to n do
        d:=d+x
print(d)
```

(B) Procedure ORD(x: int, n: positive int)

```
If n =1 then ORD(x,n):=x
else ORD(x,n):=ORD(x,n-1)+x
```

(C) Procedure ORD(n: positive int)

```
d:=0
for i = 1 to n do
        d:=d+n
print(d)
```

1) Find maximum number of vertices in a binary tree of height 4?

2) Find maximum number of vertices in a binary tree of height 5?

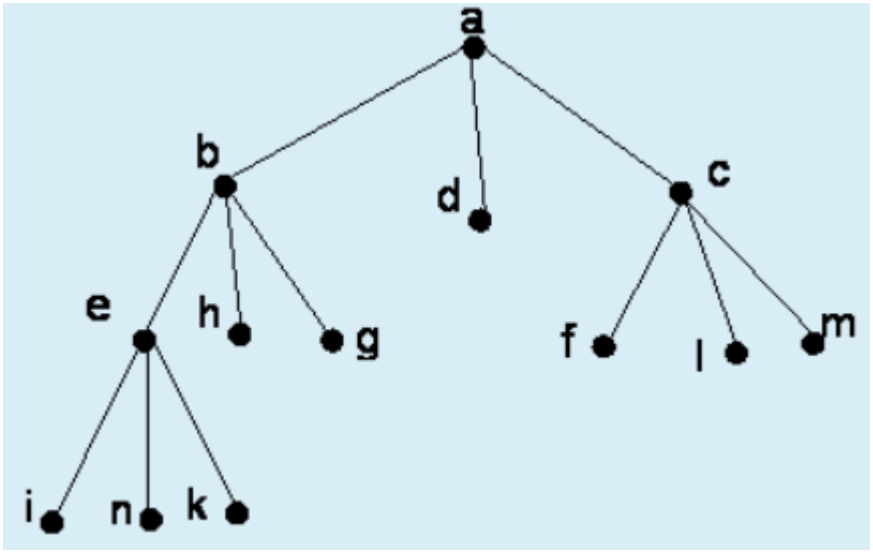
Fill in the blank: "The graph…has…edges"

(i) $C_n, 2n$

(ii) $K_{m,n}, mn$

(iii) $W_n, 2n + 1$

(iv) K_n, n^2



How many ancestors of the vertex d?

A. 2

B. 3

C. 0

D. None of the other choices is correct

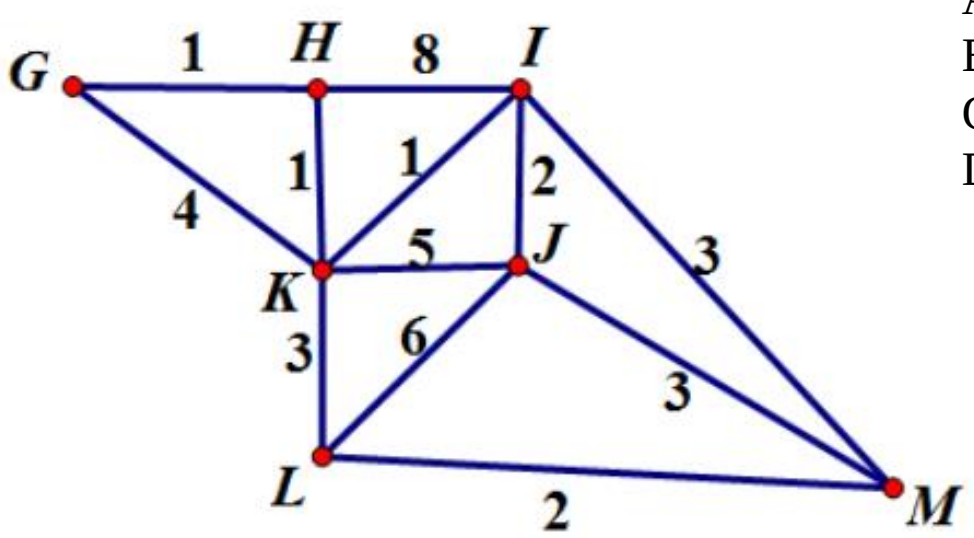
E. 1

Which of these codes are prefix codes?

A) m: 101 n: 110 p: 10 q:11
B) m: 101 n: 100 p: 11 q: 111
C) m: 101 n: 100 p: 10 q: 111
D) m: 101 n: 100 p: 110 q: 11

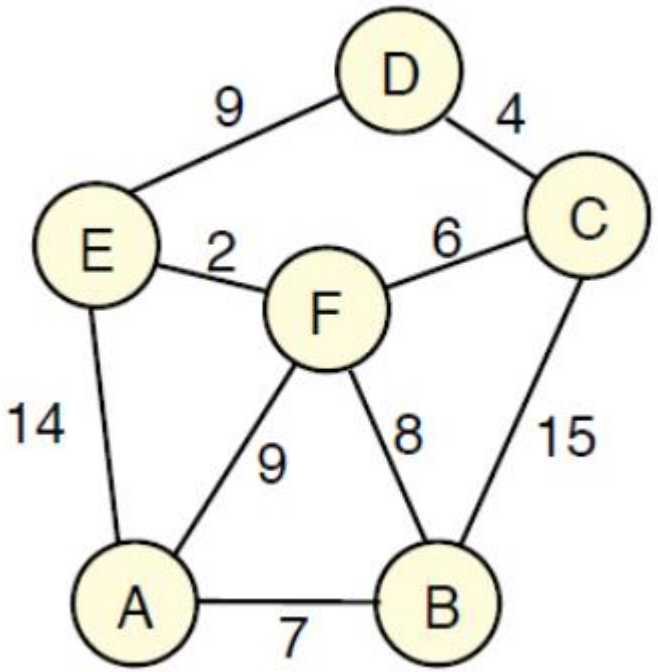
Use Huffman coding to encode these symbols with given frequencies:
A: 0.10, B: 0.25, C: 0.05, D: 0.15, E: 0.30, F: 0.07, G: 0.08.
What is the average number of bits required to encode a symbol?

What is the next vertex chosen after G, H and K when using Dijktra's algorithm to find the shortest path from G to M in this weighted graph?



A) L
B) J
C) I
D) M

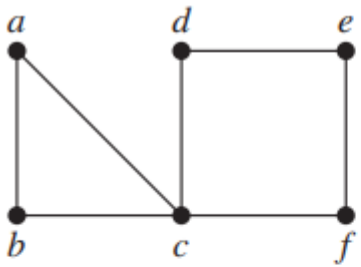
What is the next vertex chosen after A and B when using Dijktra's algorithm to find the shortest path from A to D in this weighted graph?



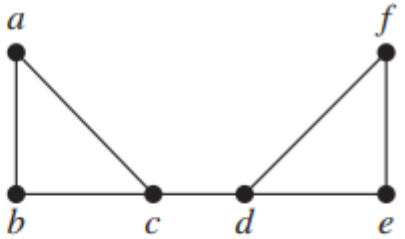
A) E
B) F
C) C
D) D

How many cut vertices and cut edges are there in each of the following graphs?

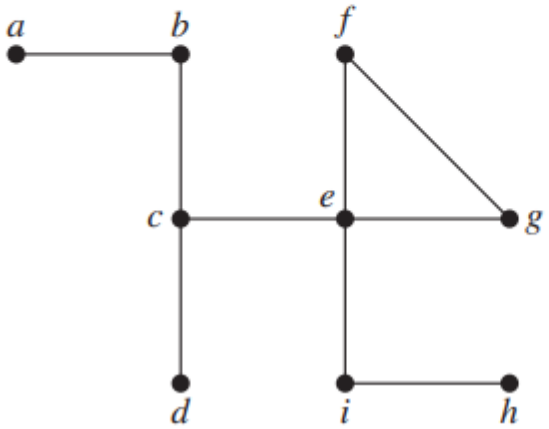
31.



32.



33.



Review:

Spanning tree and minimum spanning tree